

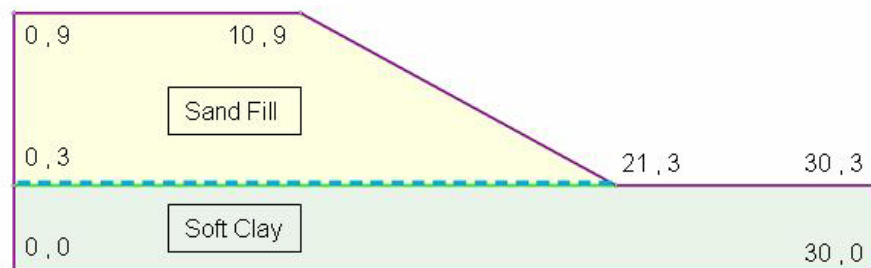
Geogrid Embankment (no slip)

This tutorial will demonstrate the use of geosynthetics in *Phase2*, by performing a shear strength reduction (SSR) analysis for a sand embankment on top of a soft clay layer with a geogrid liner in between. The geogrid liner is considered to be fully bonded to both soil layers to prevent slip at the geogrid/soil interface.

Topics Covered

- Shear strength reduction
- Slope stability
- Multiple materials
- Liner support (Geogrid)
- Graphing liner data

Geometry

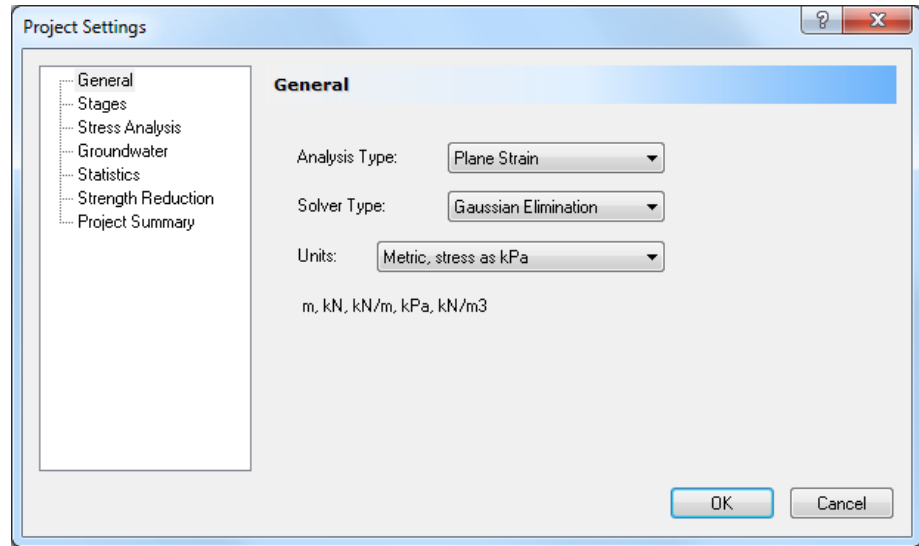


Model

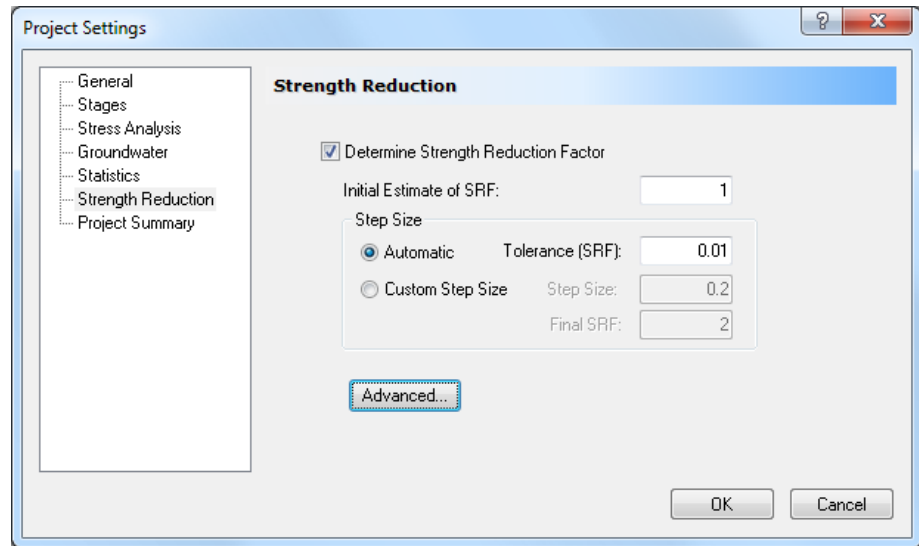
Start the *Phase2* Model program.

Project Settings

Open the Project Settings dialog from the Analysis menu and make sure the General tab is selected. Define the units as being “Metric, stress as kPa”. Do not change the number of stages and do not exit the dialog.



In the Project Settings dialog, select the Strength Reduction tab. Turn on the *Determine Strength Reduction Factor* checkbox. This enables the SSR analysis. Leave the various SSR settings at the default values. Close the Project Settings dialog by pressing the OK button.



Boundaries

This model requires an External boundary to define the geometry, and a material boundary between the sand and clay layers. Generate these boundaries as follows:

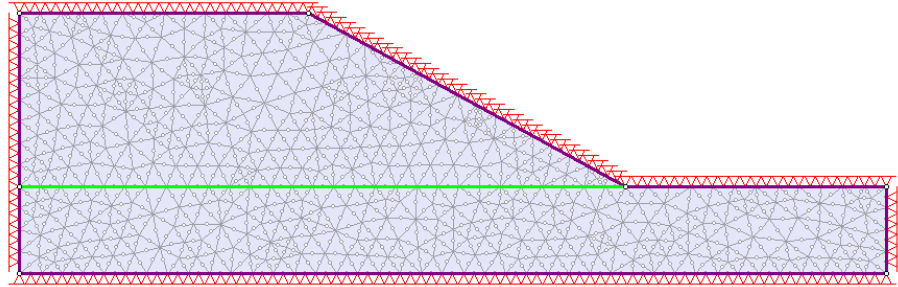
1. For the external boundary, select the **Add External** option in the Boundaries menu and enter the coordinates shown in the figure at the beginning of this tutorial.
2. For the material boundary, select the **Add Material** option in the Boundaries menu and enter the coordinates (0,3) and (21,3) or simply click on these points on the existing boundary. Hit enter to finish.

Mesh

Now generate the finite element mesh. Before we do this, let's define the parameters (type of mesh, number of elements, type of element) used in the meshing process.

1. Select the **Mesh Setup** option in the **Mesh** menu.
2. In the Mesh Setup dialog, change the Mesh Type to Uniform, the Element Type to 6 Noded Triangles and the number of elements to 800.
3. Close the Mesh Setup dialog by selecting the OK button.

Mesh the model by selecting the **Discretize and Mesh** option in the **Mesh** menu.



Mesh and default boundary conditions

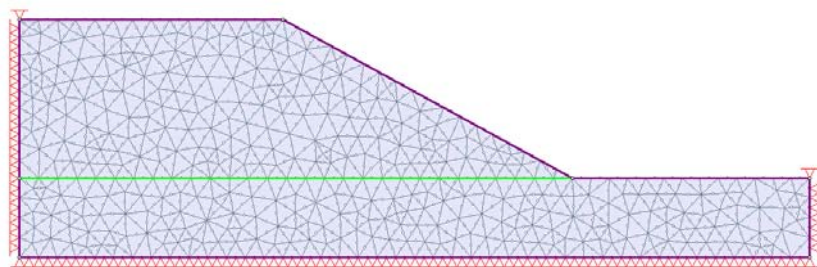
Boundary Conditions

Set the boundary conditions on the slope portion of the model. What we want to do is define the portion of the exterior boundary representing the ground surface (0,9 to 10,9 to 21,3 to 30,3) as being free to move in any direction.

1. Select the Free option in the Displacements menu.
2. Select the three line segments defining the ground surface of the slope.
3. Right-click and select the Done Selection menu option (or hit Enter).

TIP: You can also right-click on a boundary to define boundary conditions.

We now want the exterior boundaries on the left, right and bottom sides to be restrained in the x and y direction (pinned/fixed) so that these points will not move. By default these boundaries are fixed when the model is created, however freeing the surface boundaries also caused the freeing of the corner nodes (0,9 and 30,3). To re-fix these points, select the Restrain X,Y option in the Displacements menu. Select the two line segments defining the top-left and right sides, and hit enter. The following image represents what the model should now look like.



Free boundary condition applied to ground surface

Field Stress

Now define the in-situ stress field.

1. Select the **Field Stress** option in the **Loading** menu.
2. Change the Field Stress Type from Constant to Gravity (gravitational stress distribution throughout the slope).
3. Check the Use actual ground surface checkbox. By using this option, the program will automatically determine the ground surface above every finite-element and define the vertical stress in the element based on weight of the material above it.
4. Leave the horizontal stress ratios as 1, meaning hydrostatic initial stresses. If you know the horizontal stress ratio when doing your own slope model, you can use this information. However, the horizontal stress distribution within a slope is rarely known, so leaving the default hydrostatic stress field has shown to be a good assumption.

Field Stress Properties

Field Stress Type: Gravity

☒ Use actual ground surface ☐ Use effective stress ratio ☐ Use variable stress ratio

Ground Surface Elevation (m): 0

Unit Weight of Overburden (kN/m3): 27

Total Stress Ratio (horiz/vert in plane): 1

Total Stress Ratio (horiz/vert out-of-plane): 1

Locked-in horizontal stress (in plane) (kPa, Comp. +): 0

Locked-in horizontal stress (out-of-plane) (kPa, Comp. +): 0

OK Cancel Statistics... Advanced >>

Material Properties

Define the material properties of the soft clay layer and the overlying sand fill.

Select the Define Materials option in the Properties menu.

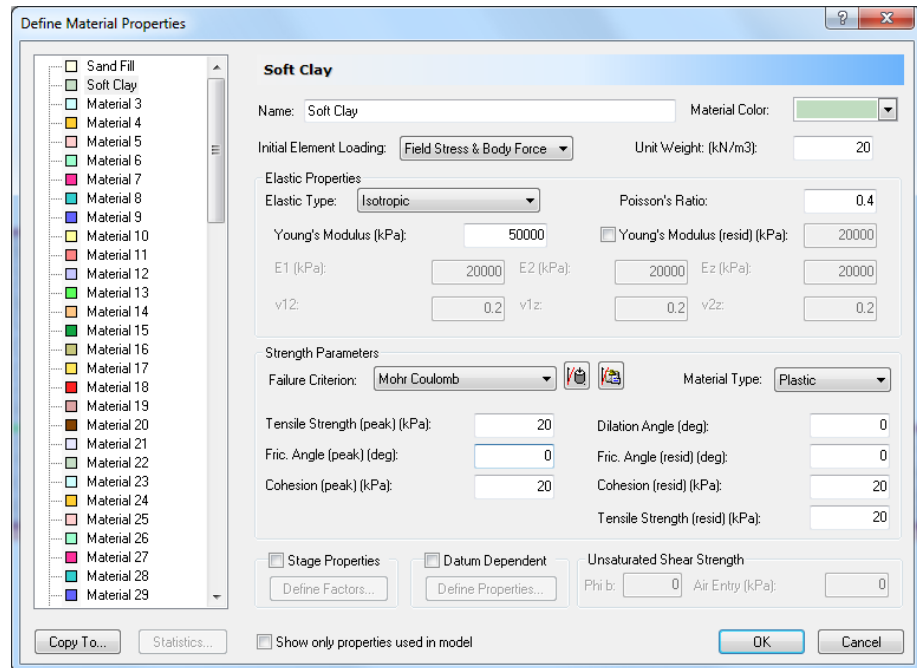
Make sure the first tab (Material 1) is selected. Type Sand Fill for the name. Change the colour to a sandy yellow if you wish. Make sure the Initial Element Loading is set to Field Stress & Body Force (both insitu stress and material self weight are applied). Enter 17 kN/m³ for the Unit Weight. For Elastic Properties, enter 50000 kPa for the Young's Modulus and 0.4 for the Poisson ratio. For Strength Parameters, make sure the Failure Criterion is set to Mohr-Coulomb. Set the Material Type to Plastic, meaning the material will yield/fail. Set the Tensile Strength, Cohesion, residual strength, and residual cohesion to zero kPa. This is representative of a dry, unconsolidated sand embankment. Set the peak and residual Friction Angle to 37°. Leave the dilation angle at 0° (no volume increase when sheared, non-associated flow rule). Do not press the OK button. The dialog should appear as shown.

The screenshot shows the 'Define Material Properties' dialog box for 'Sand Fill'. The left pane lists materials 1 through 29, with 'Sand Fill' (Material 1) selected. The main pane contains the following settings:

- Name:** Sand Fill
- Material Color:** Sandy yellow
- Initial Element Loading:** Field Stress & Body Force
- Unit Weight: (kN/m3):** 17
- Elastic Properties:**
 - Elastic Type:** Isotropic
 - Poisson's Ratio:** 0.4
 - Young's Modulus (kPa):** 50000
 - Young's Modulus (resid) (kPa):** 20000
 - E1 (kPa):** 20000
 - E2 (kPa):** 20000
 - Ez (kPa):** 20000
 - v12:** 0.2
 - v1z:** 0.2
 - v2z:** 0.2
- Strength Parameters:**
 - Failure Criterion:** Mohr Coulomb
 - Material Type:** Plastic
 - Tensile Strength (peak) (kPa):** 0
 - Dilation Angle (deg):** 0
 - Fric. Angle (peak) (deg):** 37
 - Fric. Angle (resid) (deg):** 37
 - Cohesion (peak) (kPa):** 0
 - Cohesion (resid) (kPa):** 0
 - Tensile Strength (resid) (kPa):** 0
- Stage Properties:** ☐ (unchecked)
- Datum Dependent:** ☐ (unchecked)
- Unsaturation Shear Strength:**
 - Phi b:** 0
 - Air Entry (kPa):** 0

Buttons at the bottom include 'Copy To...', 'Statistics...', 'Show only properties used in model', 'OK', and 'Cancel'.

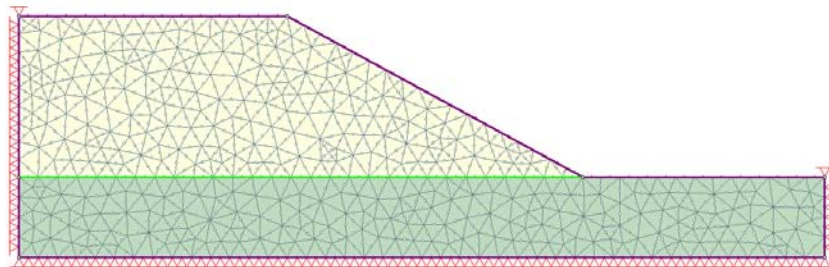
Now define the properties of the clay layer. Click on the second tab (Material 2). Change the name to Soft Clay. Enter the material properties as shown in the following diagram. Press the OK button to save the properties and close the dialog.



Assigning Properties

Assign the different material properties to different layers in the model. Select Assign Properties in the Properties menu. By default the entire model is set to Material 1 (Sand Fill). Change the bottom layer to Soft Clay by selecting Soft Clay in the dialog and clicking anywhere in the lower layer of the model. Close the dialog. The model should now appear as shown.

TIP: you can also assign material properties by right clicking in the desired material layer and choosing Assign Material from the pop-up menu.

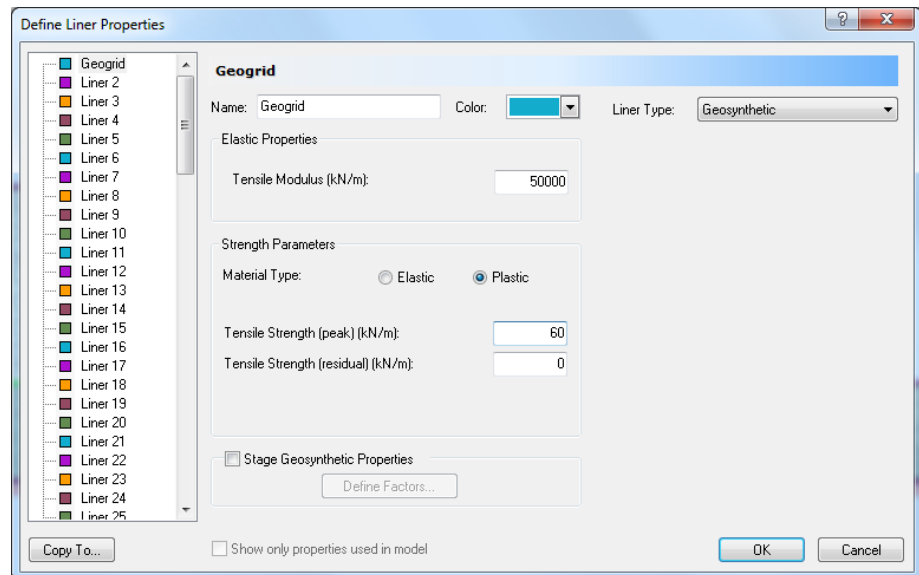


Geogrid

We now wish to add a geogrid between the clay layer and the sand fill to increase the strength of the slope. A geogrid is a flexible planar reinforcement that offers no resistance to bending or compression. The geogrid has a tensile strength only.

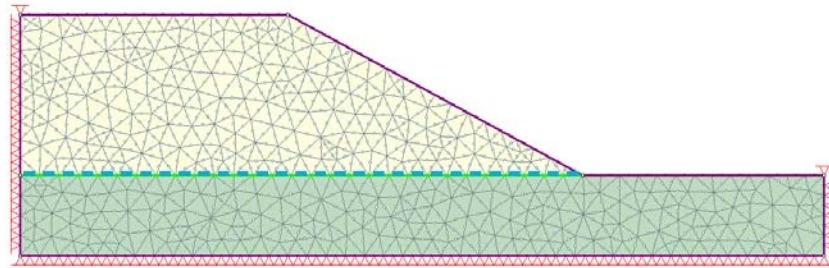
In this case the geogrid is modeled in *Phase2* as a simple liner. Note that this is only true when there is assumed to be no slip between the reinforcement layer and the soil. For geosynthetics with slip allowed see Tutorial 12 (Geogrid Embankment with Slip).

To create the geogrid in the model, first assign its properties. From the Properties menu, choose Define Liners. For Liner 1, change the name to Geogrid. At the top right of the dialog under Liner Type, select Geosynthetic. NOTE: you must change the Liner Type from Beam to Geosynthetic prior to filling in any other properties, since the geosynthetic does not possess flexural rigidity like a regular beam or liner. Now set the Tensile Modulus to 50000, set the Material Type to Plastic and set the Tensile Strength (peak) to 60. Click the OK button to save the settings and close the dialog.



Now we must add a geogrid to the model. Select Add Liner from the Support menu. Ensure that Geogrid is selected for the Liner Property. Click OK to close the box. Now click anywhere on the material boundary to install the geogrid. Right click and choose Done Selection or hit Enter to finish. Your geogrid is now installed and the model should appear as in the following figure.

NOTE: If your liner appears under the material boundary rather than above it, this is likely because you entered the points for the material boundary from right to left rather than from left to right. Although this will not affect the results, you may wish to flip the liner around so that your plots will be consistent with the plots in this tutorial. To do this, right click on the liner and choose Reverse Liner Orientation.



You have completed the definition of the model. Save the model using the Save option in the File menu.

Compute

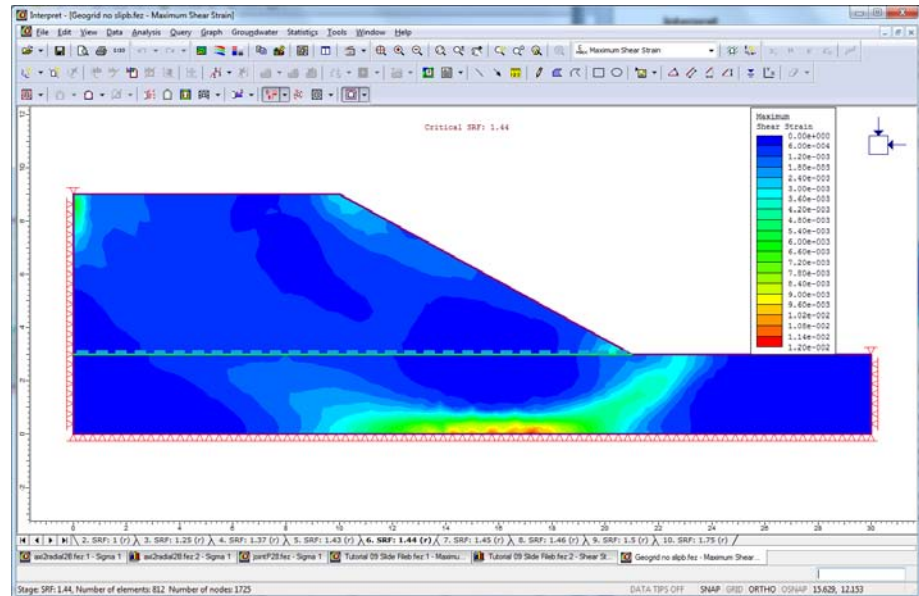
Run the model using the Compute option in the Analysis menu. The analysis should take a few minutes to run.

Once the model has finished computing (Compute dialog closes), select the Interpret option in the Analysis menu to view the results.

Interpret

After you select the Interpret option, the Interpret program starts and reads the results of the analysis. The following screen is displayed with the critical strength reduction factor (SRF) of 1.44 written at the top of the window.

TIP: Click and drag the legend to the right side of the screen so as not to cover up the model.

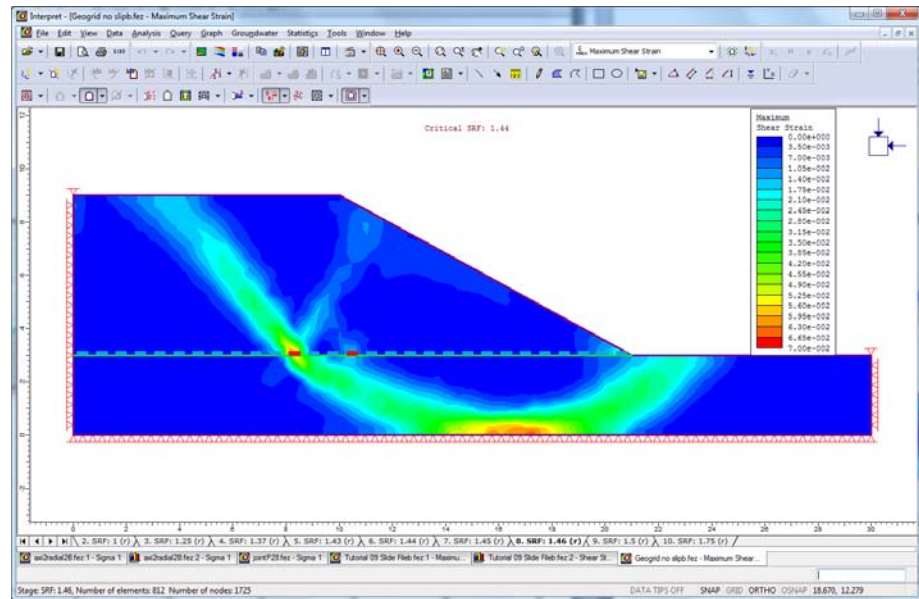


The Interpret view lists the various computed reduction factors in tabs along the bottom of the screen. The tab that is selected by default is the critical SRF. The maximum shear strain dataset is selected and contoured. Maximum shear strain will give you a good indication of where slip is occurring, especially if you change the view to higher SRF values. By cycling through the various SRF tabs, you get a good indication of the progression of failure through your slope.

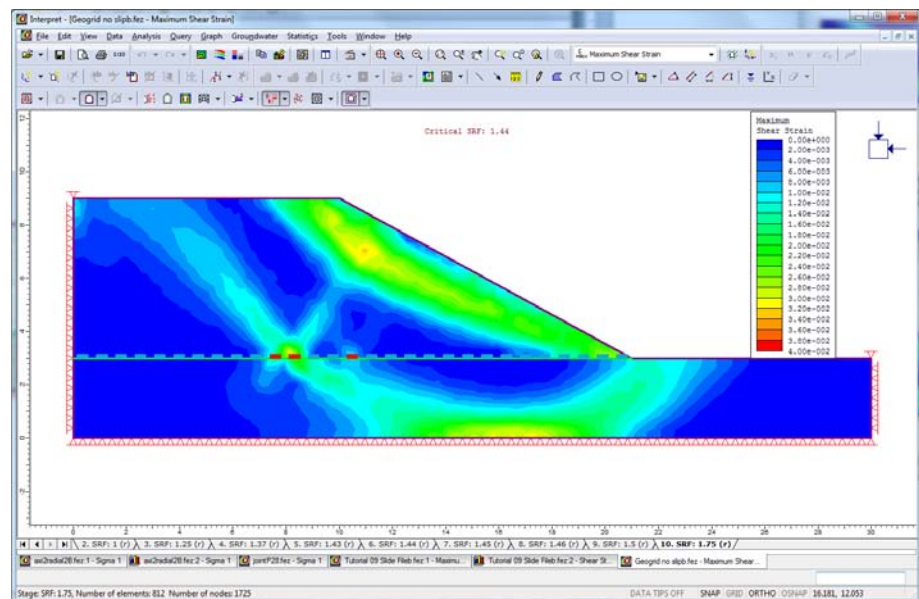


Display the yielded elements in the geogrid by clicking the Display Yielded Liners button. You should see no change, indicating that none of the geogrid elements have failed.

Change the SRF to 1.45 by clicking on the SRF: 1.45 tab at the bottom of the window. Observe that two of the geogrid elements have failed and that large shear strains accompany this failure. Clearly the tensile failure of the geogrid has resulted in unstable sliding of the slope (lack of convergence in the model).



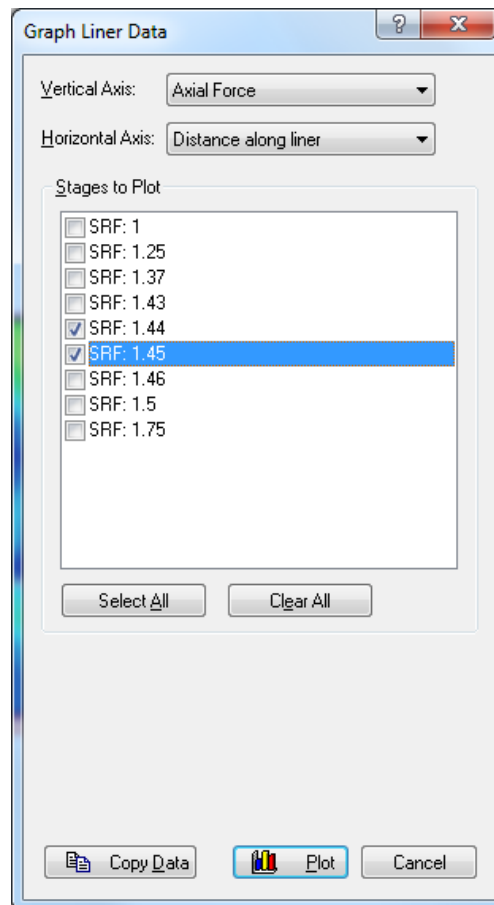
Now switch to SRF: 1.75. The view should look like the following image. Note the two well-formed shear bands in the model. Methods for further interpretation of SSR models are presented in Tutorial 8 “Shear Strength Reduction Analysis”.



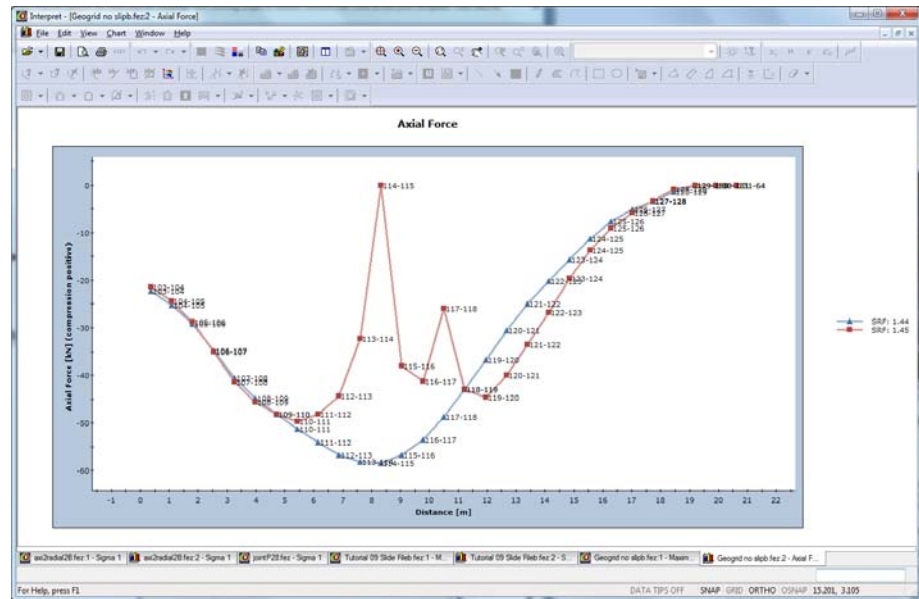
Graphing Geogrid Data

A graph of the tensile force acting on the geogrid can be easily obtained. Select the Graph Liner Data option in the Graph menu, then click on the geogrid line and hit Enter. A dialog will appear asking which data you would like to plot. Use the defaults for the Vertical Axis and Horizontal Axis. Under Stages to Plot, turn on SRF: 1.44 and SRF: 1.45. Turn off the other 'stages'. Recall that the different SRF models are considered as different stages in *Phase2*.

TIP: you can perform the same task by right clicking on the geogrid liner and choosing Graph Liner Data.



Select Plot to generate a graph of axial force along the length of the geogrid liner for the two SRF models.



This graph shows the tensile (negative) forces supported by the geogrid. Recall that the geogrid cannot support compressive forces. You can see that the failure of two elements in the SRF: 1.45 model has caused a drastic drop in the tensile forces compared with the SRF 1.44 model. This loss of support in the geogrid is what leads to the slope failure in the model.

Additional Exercise

Try re-running the model with no geogrid liner. You should observe that the critical SRF is 1.26 and that the failure surface is less localized.